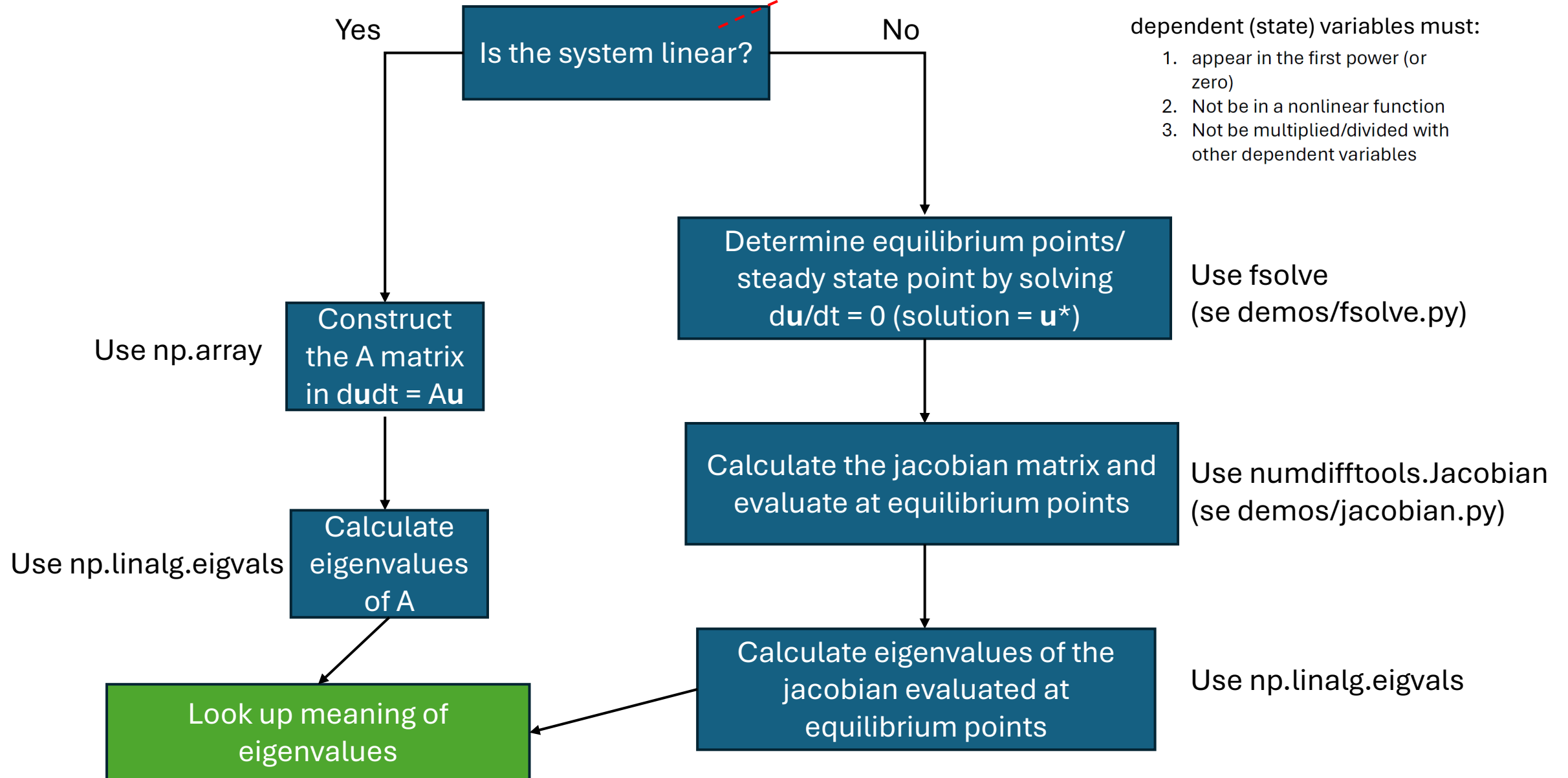


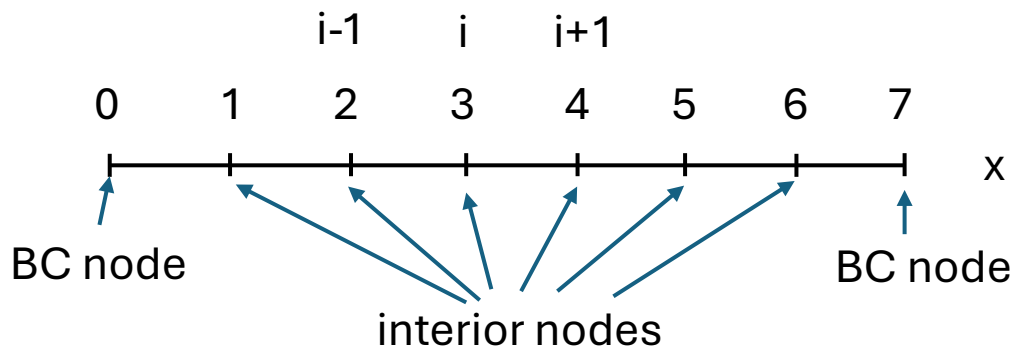
Algorithm for stability analysis



Extra hints to first exercises – you don't need **numdifftools** for the first two exercises!

1. What should k_1 and k_2 be? You can choose freely, will it affect the conclusion on stability? k_1 and k_2 are coefficients in front of state variables (so that should be reflected in the A matrix).

2. $\frac{\partial u_i}{\partial t} = D \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2}$. Is it linear? Instead of i , select some interior node that is i to make it concrete for you. The ODE in position i depends on nodes to the left and right. So you have three state variables that describe $\frac{\partial u_i}{\partial t}$.



$$\begin{matrix} & u_2 & u_3 & u_4 \\ \begin{matrix} du_2dt \\ du_3dt \\ du_4dt \end{matrix} & \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \end{matrix}$$

3. There is two equilibrium points here (see "demos/fsolve.py" on how to find them). The interesting one is NOT $[0,0]$

4. Look in stability scheme from last lecture slides (last slide)

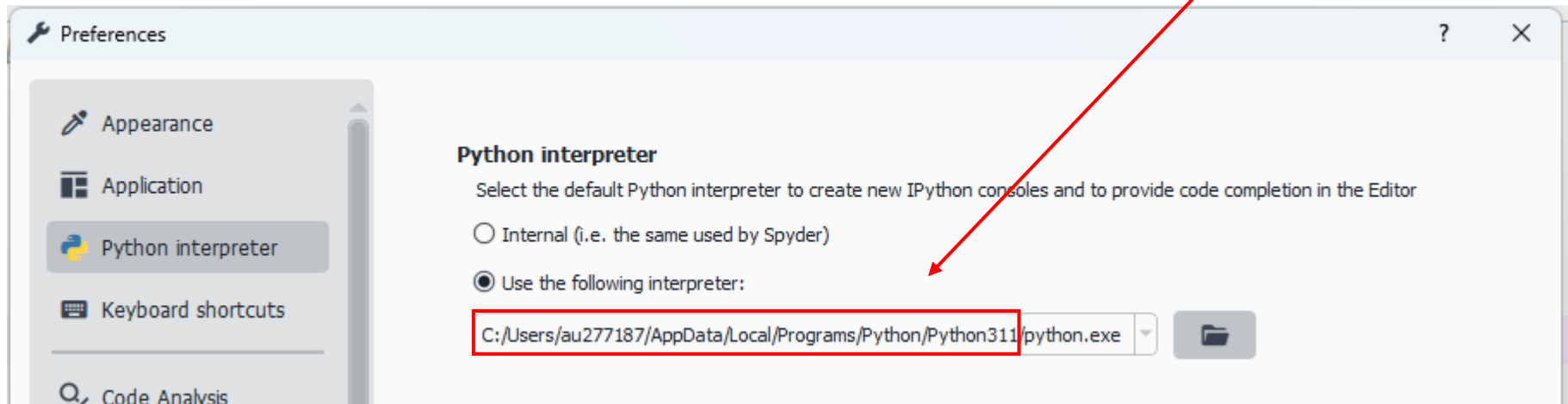
5. Recall that explicit euler formula is: $y[i + 1] = y[i] + \Delta t \cdot \frac{dy[i]}{dt}$

Spyder: numdifftools not installed!

In command prompt:

```
C:\Users\au277187>pip install numdifftools
Requirement already satisfied: numdifftools in c:\users\au277187\appdata\local\programs\python\python311\lib\site-packages (0.9.41)
```

Install location needs to match with where spyder looks



If not you have two options:

1. Change path in spyder to the one shown in the command prompt
2. Change where you install packages

Use AI (e.g.) chatgpt. Be specific about your errors and your file locations when you talk with it. Give it the path shown in spyder and in the command prompt so it can give specific answer to you.